Problem Statement

Title: Diabetic Retinopathy Detection Using Deep Learning

Introduction

Diabetic Retinopathy is a serious eye disease caused by diabetes. It damages the blood vessels of the retina and may lead to vision loss or blindness if not detected at an early stage. According to medical studies, a large number of diabetic patients suffer from this condition worldwide.

Problem Description

Traditional diagnosis of diabetic retinopathy requires examination of retinal images by experienced ophthalmologists. This manual process can be time-consuming, expensive, and may not always be accessible in rural or remote areas where medical experts are limited.

Due to the increasing number of diabetic patients, hospitals and medical centers face difficulty in screening all patients efficiently. Therefore, there is a need for an automated system that can assist doctors in detecting diabetic retinopathy quickly and accurately.

Proposed Solution

The proposed system uses **deep learning techniques** to automatically analyze retinal images and detect signs of diabetic retinopathy. A Convolutional Neural Network (CNN) model is trained using a dataset of retinal images to classify whether a patient has diabetic retinopathy or not.

The system processes the input retinal image, extracts important features using deep learning algorithms, and provides a prediction result.

Objective of the Project

- To develop an automated system for detecting diabetic retinopathy.
- To use deep learning algorithms for medical image analysis.
- To assist doctors in early diagnosis of eye diseases.
- To reduce the time required for manual screening.

Expected Outcome

The developed system will help in early detection of diabetic retinopathy, improving the chances of timely treatment and preventing blindness in diabetic patients.